

Putnam Session #1

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SIAM IIT

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What is the Putnam?

- *“The preeminent mathematics competition for undergraduate college students in the United States and Canada.”*
- Will be held on December 6th this year
- Topics from all over math
- An opportunity to solve interesting problems!

The Putnam is **hard**.

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Median score is usually 1. Test is right before finals. Why bother? You're here because you

- **Like solving problems**
- Like math
- Want to challenge yourself
- Want to meet people with similar interests

You should only do this for fun.

Still here? Let's dive in.

General information

- 2 problem sets (A and B)
- Each test lasts three hours, with a 2-hour break in between
- 6 problems each, in roughly increasing difficulty
- Each problem is worth 10 points → total of 120 points
- Topics from all over math are fair game
- Harsh grading

Today's Putnam Problem

(Putnam 2024 B1) Let n and k be positive integers. The square in the i -th row and j -th column of an n -by- n grid contains the number $i + j - k$. For which n and k is it possible to select n squares from the grid, no two in the same row or column, such that the numbers contained in the selected squares are exactly $1, 2, \dots, n$?

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Ask yourselves the following:

- Can you think of any idea to tackle the problem?
- How difficult do you perceive it now? Think of a number from 1 (approachable) to 5 (very difficult), write it down, and keep it there until the end of the session.

Plan for today

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But first, a note: all problems have an underlying “story,” a sequence of ideas they rest upon. We will be exploring

- the story of ideas behind today's problem; and
- problem-solving strategies as we explore those ideas.

Once upon a time...

A definition

A **Latin square** of order n is an $n \times n$ grid filled with n symbols (think of the numbers $1, 2, \dots, n$, such that each symbol appears exactly once in each row and column.

1	2	3	4
2	3	4	1
3	4	1	2
4	1	2	3

A Latin square of order 4.

You have already seen Latin squares and even played with them: Sudoku!

5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

How do we come up with Latin squares?

We can consider addition tables, where indices wrap around after some point $\rightarrow$ modular arithmetic!

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Our first problem!

An $n \times n$ Latin square A is *symmetric* if it is symmetric around the main diagonal, that is, $A_{ij} = A_{ji}$ for all i and j . It is *idempotent* if every symbol appears on the main diagonal. Show that if A is both symmetric and idempotent, then n is odd. Find a 5×5 symmetric, idempotent Latin square.

Our first problem!

1	2	3
2	3	1
3	1	2

1	2	3	4	5
2	4	1	5	3
3	5	4	2	1
4	1	5	3	2
5	3	2	1	4

Figure: Example for $n = 3$, non-example for $n = 5$.

Solution

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- How many 1's are there above the main diagonal?

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- How many 1's are there above the main diagonal? **k**
- How many 1's are there on the main diagonal?

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On one hand, we know there are n 1's on the square.

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- How many 1's are there above the main diagonal? k
- How many 1's are there on the main diagonal? 1
- How many 1's are there in total?

Solution

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On one hand, we know there are n 1's on the square.

On the other hand, let k be the number of 1's below the main diagonal.

- How many 1's are there above the main diagonal? **k**
- How many 1's are there on the main diagonal? **1**
- How many 1's are there in total? **$2k + 1$**

Then $n = 2k + 1$, i.e., n is odd.

Solution

How can we generate a symmetric, idempotent Latin square of order 5?

Solution

How can we generate a symmetric, idempotent Latin square of order 5? Addition table modulo 5!

0	1	2	3	4
1	2	3	4	0
2	3	4	0	1
3	4	0	1	2
4	0	1	2	3

This works for every odd $n \rightarrow$ we'll leave it to you to prove this.

The plot thickens...

Another puzzle about Latin squares

A *transversal* on an $n \times n$ Latin square is a set of n cells, no two in the same row or column, such that there is a cell containing each symbol.

1	2	3	4	5
2	4	1	5	3
3	5	4	2	1
4	1	5	3	2
5	3	2	1	4

Figure: Transversal on 5×5 Latin square.

Our second problem!

Note that the addition table modulo n is a Latin square.

0	1	2	3
1	2	3	0
2	3	0	1
3	0	1	2

0	1	2	3	4
1	2	3	4	0
2	3	4	0	1
3	4	0	1	2
4	0	1	2	3

For which integers $n \geq 2$ can we find a transversal in the addition table modulo n ?

Interlude: problem-solving strategies

So far, we've been following two “modes” of thinking:

- Exploration: trying out some examples and see if they work.
- Restriction: ruling out cases from your search or, even better, prove they don't work.

Balancing the two is key.

Idea of solution

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Conjecture: we can only find a transversal when n is **odd**.

Solution requires showing two things:

- Show that no transversal exists when n is even;
- Construct a transversal when n is odd.

Solution

Suppose n is even, and assume for the sake of contradiction that there exists a transversal of the addition table modulo n . What does that mean?

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What does that mean?

Impose a coordinate system on the grid, where (i, j) is the cell in the i -th row and j -th column. Let $(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n)$ be the cells of the transversal. What does this mean about these numbers?

0	1	2	3
1	2	3	0
2	3	0	1
3	0	1	2

(0, 0)	(0, 1)	(0, 2)	(0, 3)
(1, 0)	(1, 1)	(1, 2)	(1, 3)
(2, 0)	(2, 1)	(2, 2)	(2, 3)
(3, 0)	(3, 1)	(3, 2)	(3, 3)

Solution

Recall that $(a_1, b_1), \dots, (a_n, b_n)$ are the cells of the transversal.
Which numbers are contained in them?

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Which numbers are contained in them?

$$(a_1, b_1) \rightarrow a_1 + b_1 \pmod{n}, \dots, (a_n, b_n) \rightarrow a_n + b_n \pmod{n}$$

Solution

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$$(a_1, b_1) \rightarrow a_1 + b_1 \pmod{n}, \dots, (a_n, b_n) \rightarrow a_n + b_n \pmod{n}$$

Then the numbers $a_1 + b_1, \dots, a_n + b_n$ are
exactly the numbers $0, 1, \dots, n-1$ modulo n .

Solution

Since n is even, the sum of $0, 1, \dots, n - 1$ is congruent to $\frac{n}{2} \rightarrow$ something that is not 0.

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However, recall that $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are the numbers $0, 1, \dots, n-1$ in some order. Therefore,

Solution

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However, recall that $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are the numbers $0, 1, \dots, n-1$ in some order. Therefore,

$$(a_1 + b_1) + \dots + (a_n + b_n) \equiv \frac{n}{2} + \frac{n}{2} \equiv 0 \pmod{n}.$$

Contradiction!

Solution

The n odd case I leave to you. But it's a summary of our discussion so far: just choose the main diagonal and prove that it contains all numbers exactly once.

Showtime!

Putnam 2024, B1

Let n and k be positive integers. The square in the i -th row and j -th column of an n -by- n grid contains the number $i + j - k$. For which n and k is it possible to select n squares from the grid, no two in the same row or column, such that the numbers contained in the selected squares are exactly $1, 2, \dots, n$?

Putnam 2024, B1

Does the definition of the board not remind you of something?

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Does the definition of the board not remind you of something?
Modulo n , it's *almost* the addition table modulo n . It is the addition table, but with numbers rotated around.

-1	0	1	2	3		4	0	1	2	3
0	1	2	3	4		0	1	2	3	4
1	2	3	4	5	→	1	2	3	4	0
2	3	4	5	6		2	3	4	0	1
3	4	5	6	7		3	4	0	1	2

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And what about the set of squares we need to choose?

Putnam 2024, B1

And what about the set of squares we need to choose?

It is a transversal on the *almost* addition table! And we know for which values of n we can find a transversal on the addition table modulo n .

-1	0	1	2	3
0	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7

→

4	0	1	2	3
0	1	2	3	4
1	2	3	4	0
2	3	4	0	1
3	4	0	1	2

Solution idea

Therefore, we have the following idea:

- Prove that we can't find the construction we need when n is even.
- When n is odd, there may or may not be a construction for some values of $k \rightarrow$ we'll find that out later.

Solution

Impose the same coordinate system as before, but with numbers from 1 to n instead of 0 to $n - 1$. Let $(a_1, b_1), \dots, (a_n, b_n)$ be cells of the grid, where no two are in the same row or column.

Solution

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Then $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are the numbers $1, \dots, n$ in some order. What are the numbers in each cell?

Solution

Impose the same coordinate system as before, but with numbers from 1 to n instead of 0 to $n - 1$. Let $(a_1, b_1), \dots, (a_n, b_n)$ be cells of the grid, where no two are in the same row or column.

Then $a_1, \dots, a_n$ and $b_1, \dots, b_n$ are the numbers $1, \dots, n$ in some order. What are the numbers in each cell?

$$a_1 + b_1 - k, \dots, a_n + b_n - k.$$

Solution

If the numbers in these n cells in fact happen to be $1, \dots, n$ in some order, then

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$$\begin{aligned}\frac{n(n+1)}{2} &= \sum_{i=1}^n (a_i + b_i - k) \\ &= \sum_{i=1}^n a_i + \sum_{i=1}^n b_i - \sum_{i=1}^n k \\ &= \frac{n(n+1)}{2} + \frac{n(n+1)}{2} - nk.\end{aligned}$$

Solution

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Then $k = \frac{n+1}{2}$. Not only we found that n must be odd, but we also found the only value of k that *may* work!

Solution

Now we need to find a construction when n is odd and $k = \frac{n+1}{2}$.

-1	0	1	2	3
0	1	2	3	4
1	2	3	4	5
2	3	4	5	6
3	4	5	6	7

What pattern can we derive from this construction?

Solution

“Upper diagonal”:

$$\left(1, \frac{n+1}{2}\right), \left(2, \frac{n+3}{2}\right), \dots, \left(\frac{n+1}{2}, n\right)$$

“Lower diagonal”:

$$\left(\frac{n+3}{2}, 1\right), \left(\frac{n+5}{2}, 2\right), \dots, \left(n, \frac{n-1}{2}\right)$$

Convince yourselves that this configuration works.

Epilogue

What next?

Yay! We solved the problem! Before moving on, ask yourselves:

- How difficult do you perceive it now? Rank it on a scale from 1 to 5, as before. What is the difference compared to the ranking you made earlier?
- Can you think of another way to approach the problem?
- Overall, can you write the solution on your own? Putnam grading is **harsh** → you'd better write well.

What next?

This is just the beginning: there are many more problems waiting to be explored.

And once again, you **should only do this for fun** → median score is ≈ 1 !

Further resources

Feel free to email me! `lmarciaga@hawk.illinoistech.edu`
You can find all of us here every Wednesday at the same time and place as today.

Thanks for coming!